TRAFFIC DECISION LAB / EARLY EXPERIMENT

Can models make useful signal decisions?

Five controllers. Identical demand. One shared planning task.

Same input format

NS / EW queues
Arrival forecast
Signal phase + timers
Legal actions

→

Five controllers

Gurobi
Microsoft · Jev · Luna
Sol, medium reasoning
Choose NS or EW

→

Same rules

15 s planning horizon
120 s SUMO episode
5 / 3 / 2 s signal timing
78 cars · demand seed 7

TRAFFIC OUTCOMES

Completion, all-demand system time, travel / wait, approach coverage.

States diverge after different actions. Optimization quality is evaluated separately on frozen instances.

No new frozen optimization benchmark has been run for the revised objective.